

CIS 6270

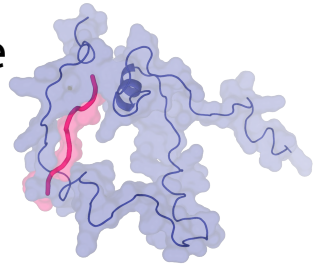
Module 1: Introduction and Math Foundations

Lecture 1.2: The Probability Simplex and Linear Algebra

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Can we do this for **finite** probability vectors?

Let X_0 be a clean sample and X_t be its corrupted version at time t.

The forward corruption distribution is: $p(x_t \mid x_0)$

The reverse denoising distribution is: $p(x_0 \mid x_t)$

The noisy marginal is

$$p(x_t) = \int p(x_t \mid x_0) p(x_0)\, dx_0$$

The noisy distribution $p(x_t)$ combines every possible clean sample x_0 with the noise process $p(x_t|x_0)$. During training, we approximate this automatically by sampling x_0 from the dataset and corrupting it to obtain x_t.

A denoising model approximates information about this reverse conditional distribution.

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The vector p contains K probabilities, and p_i is the probability that X takes the state x_i.

A valid probability vector satisfies

$$p_i \geq 0 \qquad \text{for every } i$$

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Every valid probability vector lives in the probability simplex.

The K-state probability simplex

$$\Delta^{K-1} = \left\{ p \in \mathbb{R}^K \;\middle|\; p_i \geq 0, \quad \sum_{i=1}^{K} p_i = 1 \right\}.$$

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The simplex contains every possible probability distribution over K states.

Why does the simplex have dimension K-1?

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Once the first K-1 probabilities are chosen, the final probability is fixed. Therefore, the simplex has dimension K minus one.

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the final coordinate must satisfy

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You can freely choose K-1 probabilities; the last must fill the remainder to one!

Simplex Geometry

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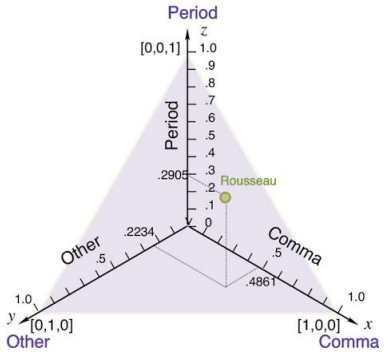
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For four states, Δ^3 is a filled tetrahedron.

Simplex Geometry

The vertices correspond to distributions that place all probability on one state

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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Points near a vertex strongly favor one state. Points near the center distribute probability more evenly.

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$$p \in \Delta^2$$

The probability vector p belongs to the two-dimensional simplex Δ^2 because its three entries are nonnegative and sum to one.

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The expected value of h(X) equals the dot product between the probability vector p and the property vector h.

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$$\mathbb{E}[h(X)] = (0.5)(1) + (0.3)(0) + (0.2)(4) = 1.3$$

1.3 is the average across many draws from the three states; any single draw still has value 1, 0, or 4.

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$$\begin{aligned} \sum_i r_i &= \lambda \sum_i p_i + (1 - \lambda) \sum_i q_i \\ &= \lambda + (1 - \lambda) \\ &= 1. \end{aligned}$$

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If p = (1, 0), q = (0, 1), and λ = 0.3, then r = 0.3p + 0.7q = (0.3, 0.7), whose entries sum to 1.

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Therefore,

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The line segment between any two points in the simplex remains inside the simplex.

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This property becomes important when interpolating between probability distributions.

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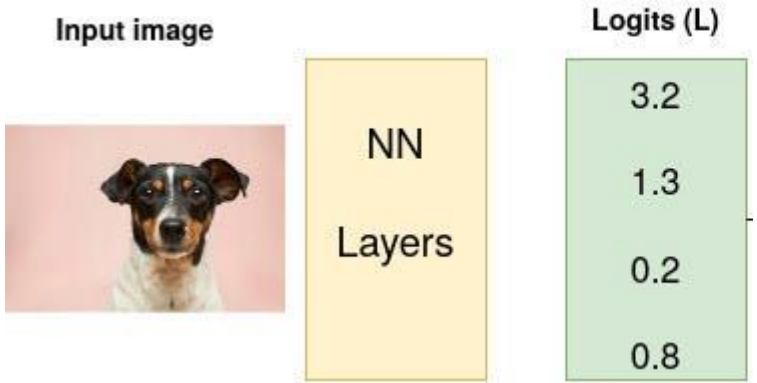
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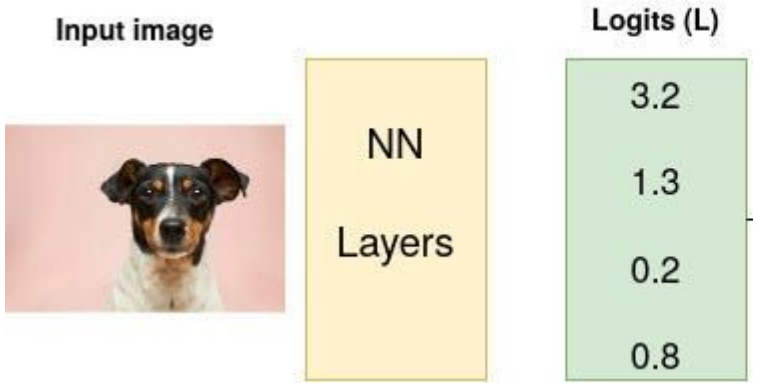
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But remember, a neural network often produces unrestricted real numbers called logits



We need to bring these values back on the simplex.



We can take the Softmax

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The probabilities sum to one:

$$\sum_{i=1}^{K} p_i = \frac{\sum_i \exp(z_i)}{\sum_j \exp(z_j)} = 1$$

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Those final values sum to 1!

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Neural models avoid storing a separate probability for every complete sequence. They calculate structured conditional or denoising distributions instead.

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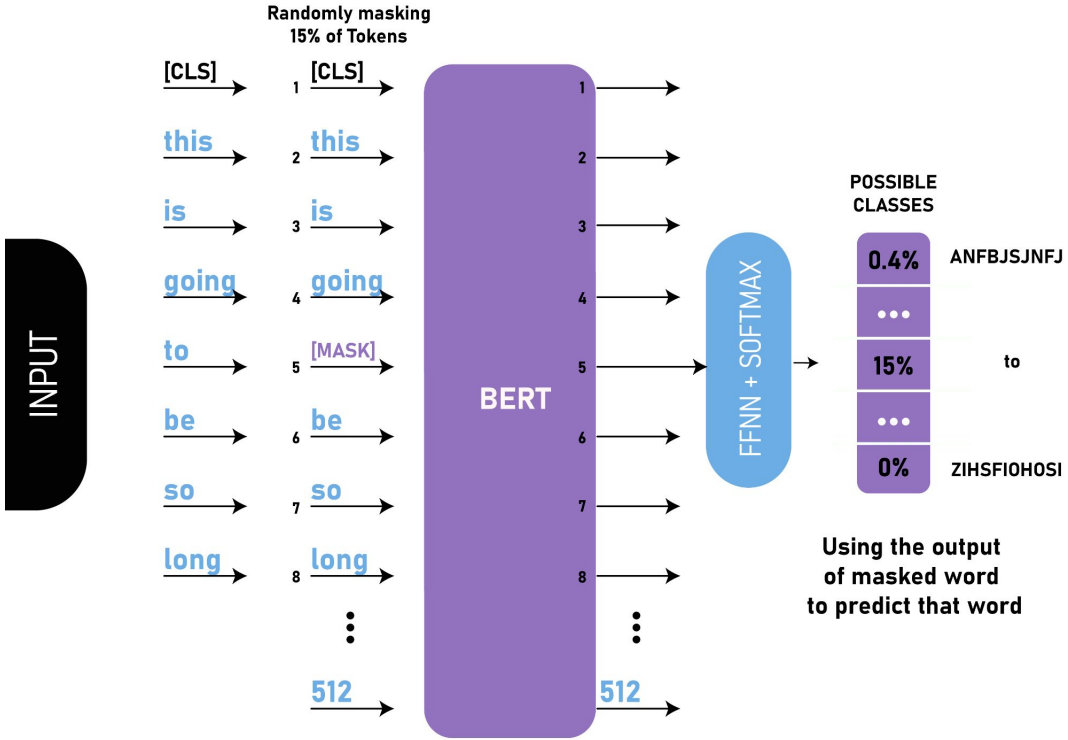
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So you only compute the probability of each nucleotide from its context when needed. This represents the enormous sequence distribution implicitly

This is essentially how you train foundation models.



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Let’s review a few vector/matrix transformations.

Vector addition and scaling

For vectors x and y,

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A linear combination has the form

$$\alpha x + \beta y$$

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A positive value indicates similar directions, a negative value indicates opposing directions, and zero indicates orthogonality.

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You'll see this in flow-matching losses, Gaussian densities, and reconstruction losses.

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Matrices

A matrix with m rows and d columns is:

$$A = \begin{bmatrix} A_{11} & \cdots & A_{1d} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{md} \end{bmatrix} \in \mathbb{R}^{m \times d}$$

Matrices

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$$A = \begin{bmatrix} A_{11} & \cdots & A_{1d} \\ \vdots & \ddots & \vdots \\ A_{m1} & \cdots & A_{md} \end{bmatrix} \in \mathbb{R}^{m \times d}$$

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Component-wise,

$$y_i = \sum_{j=1}^{d} A_{ij} x_j$$

An example

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Columns reveal the transformation

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These are exactly the columns of A.

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$$\begin{aligned} Ax &= Ae_1 + 2Ae_2 \\ &= \begin{bmatrix} 2 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 6 \end{bmatrix}. \end{aligned}$$

A matrix is completely determined by where it sends vectors. $A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

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Basically, a linear transformation leaves the origin (0,0,...,) fixed.

A short example

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The left side is:

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The right side is

$$\begin{aligned} 2Ax + Ay &= 2 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 3 \end{bmatrix}. \end{aligned}$$

Both sides agree, confirming linearity.

Let

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$$\begin{aligned} y &= Ax + b \\ &= \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ 3 \end{bmatrix}. \end{aligned}$$

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But order matters:

$$BA \neq AB$$

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Invertibility means that every output has exactly one corresponding input.

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The translation b changes the mean and leaves the covariance unchanged. The matrix A changes the directions and magnitudes of variation.

Short example

$$\mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

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The first coordinate now has variance four. The second coordinate retains variance one.

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Each new probability is the sum of probability arriving from every previous state.

Consider two states and the transition matrix

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Then

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The next-step probability of state one is 0.55, and the next-step probability of state two is 0.45

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The probabilities remain nonnegative and sum to one.

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This matrix update will be very important for Markov chains, discrete diffusion, and continuous-time jump processes.

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But what do repeated matrix transformations do over time?

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Directly calculating large powers of A can be difficult.

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Eigenvectors identify directions where matrix multiplication becomes simple scalar multiplication.

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Consider $$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Multiplying v_1 by A produces three times v_1, so v_1 is an eigenvector with eigenvalue three.

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The direction v_1 grows by a factor of three during every update, while the direction v_2 remains unchanged.

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$$v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

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Geometrically, v_1 points along the diagonal where both coordinates increase together. The direction v_2 points along the opposite diagonal where one coordinate increases while the other decreases.

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A nonzero solution v exists only when A - λI is singular:

$$\boxed{\det(A - \lambda I) = 0}$$

The eigenvalues λ are the values that make the determinant of A - λ * the identity matrix = zero.

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The characteristic equation det(A – λI) = 0 is the equation whose solutions λ are the eigenvalues of A.

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$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

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$$\begin{aligned} \det(A - \lambda I) &= (2-\lambda)^2 - 1 \\ &= \lambda^2 - 4\lambda + 3 \\ &= (\lambda - 1)(\lambda - 3) \\ &= 0. \end{aligned}$$

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Any nonzero scalar multiple of an eigenvector represents the same eigenvector direction.

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Repeated matrix multiplication can create complicated coordinate interactions.

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Repeated matrix multiplication can create complicated coordinate interactions.

Along an eigenvector direction, the behavior reduces to repeated multiplication by one number.

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$$A^n v_1 = 3^n v_1$$

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The v_1 component eventually dominates because 3^n grows while 1^n remains constant.

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$$= \begin{bmatrix} 5 \\ 4 \end{bmatrix}.$$

Each eigenvector component evolves independently according to its eigenvalue.

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Multiplication by V converts the result back to the original coordinate system.

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The n-th power of A is obtained by raising each eigenvalue to the n-th power while preserving the eigenvector directions.

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This explains why repeated transitions can forget their initial conditions and converge toward a dominant direction.

There are stationary distributions

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A stationary distribution p* satisfies

$$P^\top p^* = p^*$$

The stationary distribution p* remains unchanged after multiplication by $P^\top$.

This is an eigenvector equation with eigenvalue one:

$$P^\top p^* = 1 \cdot p^*$$

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Checking the result,

$$P^\top p^* = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = p^*$$

The distribution remains unchanged under further transitions.

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$$

The stationary distribution satisfies

$$\begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix} \begin{bmatrix} p_1^* \\ p_2^* \end{bmatrix} = \begin{bmatrix} p_1^* \\ p_2^* \end{bmatrix}$$

The first equation gives

$$0.8p_1^* + 0.3p_2^* = p_1^* \qquad 0.3p_2^* = 0.2p_1^*$$

Together with:

$$p_1^* + p_2^* = 1 \qquad p^* = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

Checking the result,

$$P^\top p^* = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} = p^*$$

We can move from discrete steps to continuous time

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Remember, the exponential function reproduces itself when differentiated:

$$\frac{d}{dt}e^{at} = ae^{at}$$

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The state x_t equals the matrix exponential e raised to tA multiplied by the initial state x_0.

The matrix exponential is defined by the power series

$$e^{tA} = I + tA + \frac{t^2A^2}{2!} + \frac{t^3A^3}{3!} + \cdots$$

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So applying the transformation once, twice, and three times, with the factorials making higher-order contributions progressively smaller…

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The full series accumulates those infinitesimal updates over time.

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$$x_{t=0} = e^{0A}x_0 = Ix_0 = x_0$$

The matrix exponential satisfies both the differential equation and the initial condition.

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The exponential of the diagonal matrix is

$$e^{t\Lambda} = \begin{bmatrix} e^{t\lambda_1} & 0 & \cdots & 0 \\ 0 & e^{t\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{t\lambda_d} \end{bmatrix}$$

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The matrix exponential scales each eigenvector direction by e^t times its corresponding eigenvalue.

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Each eigenvector direction evolves independently and continuously.

There’s stability in continuous time

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- A negative real eigenvalue produces exponential decay.
- A zero eigenvalue produces a constant component.
- A positive real eigenvalue produces exponential growth.
- A complex eigenvalue produces oscillation with growth or decay determined by its real part.

For example,

$$\lambda = -2$$

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Gives

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This is the continuous-time meaning of stability.

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Remember a covariance matrix Σ is symmetric and positive semidefinite. It can be diagonalized as:

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For a two-dimensional Gaussian, one eigenvector points along the long axis of the density ellipse, and the other points along the short axis.

You’ll see these ideas a lot!

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But we can't just do linear transformations

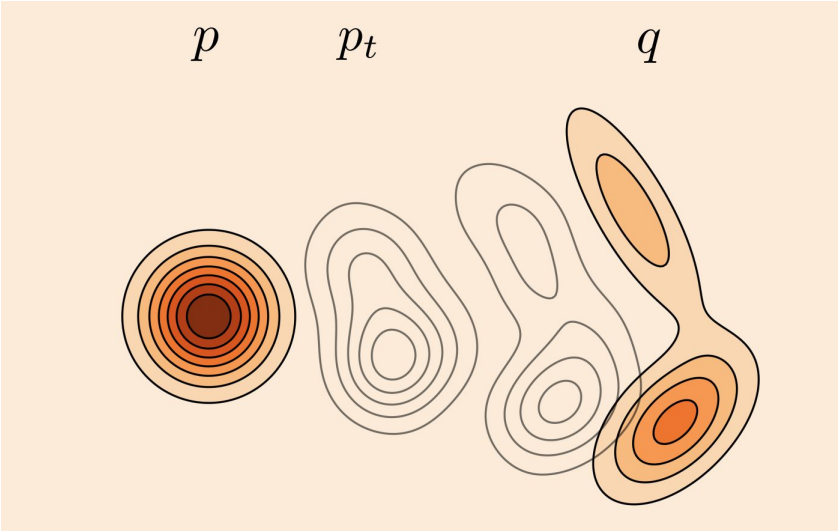
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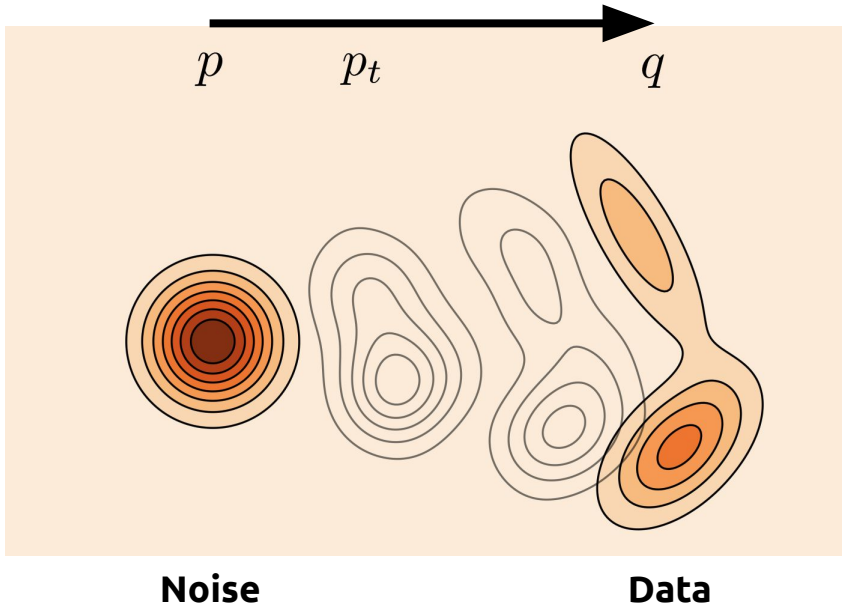
We need general nonlinear maps.

What is a generative model?



Remember what a generative model is.

- A learned process that transforms a SIMPLE distribution to the DATA distribution.



$$p_0(\mathbf{x}) \xrightarrow{p_t(\mathbf{x})} q(\mathbf{x})$$

To understand this, we need to separate two related effects:

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- A map moves individual points.

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- A map moves individual points.
- Applying the map to a random variable changes its probability distribution.

Maps

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A map from a source space $\mathscr{X}$ to a target space $\mathscr{Y}$ is written as

$$\phi : \mathcal{X} \to \mathcal{Y}, \qquad y = \phi(x)$$

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The map ϕ sends an input state x from the source space $\mathscr{X}$ to an output state y in the target space $\mathscr{Y}$, where y equals ϕ applied to x.

An example

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Let

$$\phi(x) = 2x + 3$$

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Then

$$\phi(1) = 2(1) + 3 = 5$$

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and

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And

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The map scales each input by two and shifts it by three.

Multidimensional maps

A map can transform vectors

$$\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$$

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It may be written component-wise as

$$\phi(x) = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \\ \vdots \\ \phi_m(x) \end{bmatrix}$$

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The map ϕ sends a d-dimensional input vector x to an m-dimensional output vector whose components are the functions ϕ_1 through ϕ_m.

Here’s an example

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$$\phi(x_1, x_2) = \begin{bmatrix} x_1 + x_2 \\ x_1 x_2 \end{bmatrix}$$

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$$\phi(2, 3) = \begin{bmatrix} 2 + 3 \\ 2(3) \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

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This is a nonlinear map because its second component contains the product $x_1 x_2$.